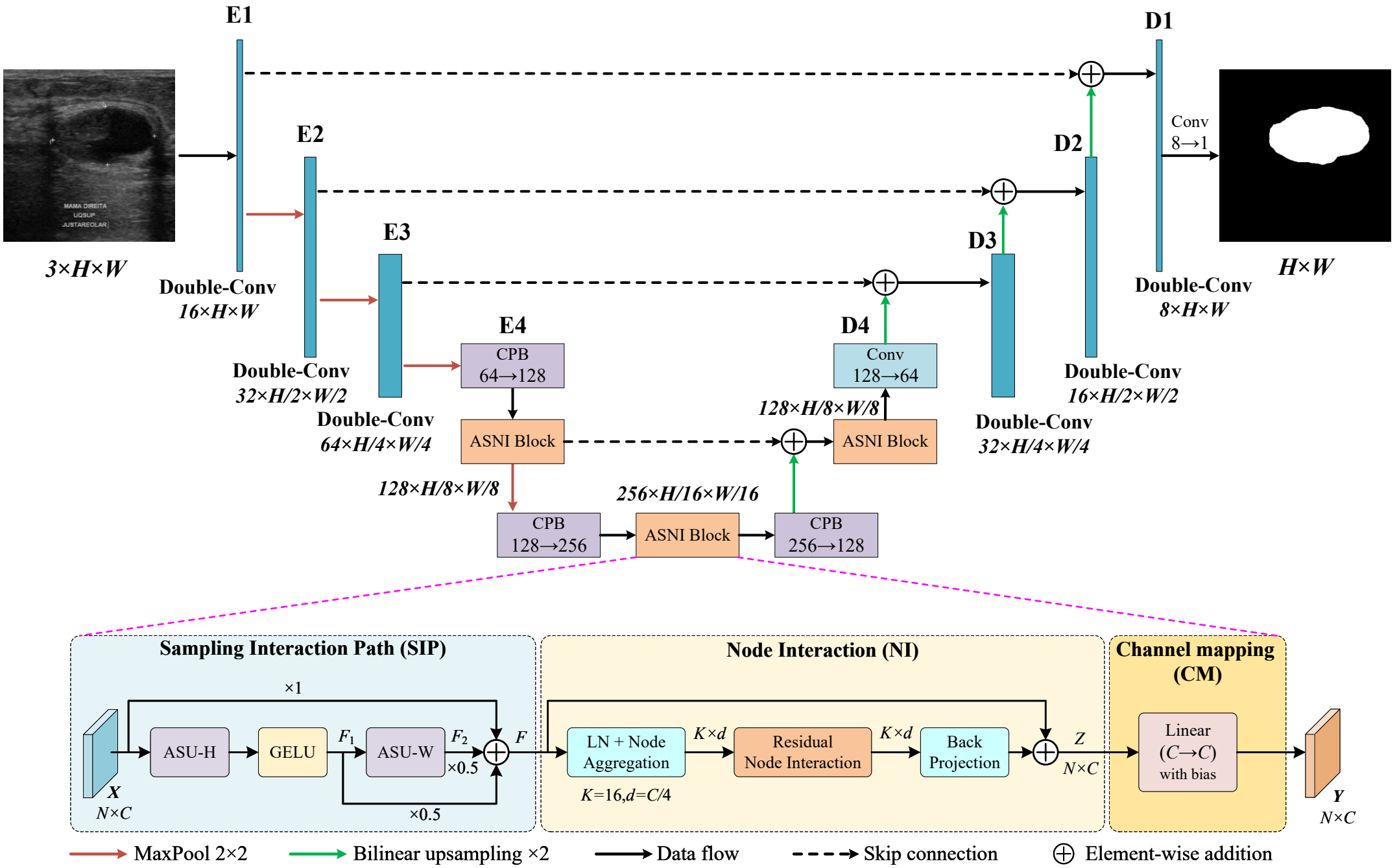

E1
E2
E3
E4
D1
D2
D3
D4
3×H×W
Double-Conv
16×H×W
Double-Conv
32×H/2×W/2
Double-Conv
64×H/4×W/4
CPB
64→128
ASNI Block
128×H/8×W/8
CPB
128→256
ASNI Block
256×H/16×W/16
CPB
256→128
ASNI Block
128×H/8×W/8
Conv
128→64
Double-Conv
32×H/4×W/4
Double-Conv
16×H/2×W/2
Double-Conv
8×H×W
Conv
8→1
H×W
Sampling Interaction Path (SIP)
×1
ASU-H
GELU
F1
ASU-W
F2
×0.5
F
×0.5
X
N×C
Node Interaction (NI)
LN + Node
Aggregation
K×d
Residual
Node Interaction
K×d
Back
Projection
Z
N×C
K=16,d=C/4
Channel mapping
(CM)
Linear
(C→C)
with bias
Y
N×C
MaxPool 2×2
Bilinear upsampling ×2
Data flow
Skip connection
Element-wise addition